MIL-S-7886/1004B(AV)
14 October 1975
Superseding
MIL-S-7886/1004A(AV)
14 October 1970

MILITARY SPECIFICATION

SPARK PLUG: SHIELDED, FINE WIRE PLATINUM ELECTRODE, TYPE 4 FOR AIRCRAFT RECIPROCATING ENGINES

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers type 4, fine wire platinum electrode spark plug for:

Aircraft	Using Engine
U-1A	R-1340-59 and R-1340-61
O-1A, O-1D, -1E, and -1G	O-470-11 and O-470-15
U-6A	R-985-AN-39A
U-8D, U-8F, and U-8G	O-480-1A, -1B and O-480-3, -3A
OH-13H	O-435-23C

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

P-D-680	Dry Cleaning Solvent

MILITARY

MIL-L-6081	Lubricating Oil, Jet Engine
MIL-S-7886	Spark Plug, Shielded, Aircraft Reciprocating Engine, General Specification for

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STANDARDS

MILITARY

MIL-STD-414	Sampling Procedure and Table for Inspection by Variables for Percent Defective
MS17840	Gasket — Spark Plug

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Qualification.—Spark plug shall be a product which has passed the qualification tests in accordance with MIL-S-7886; which has been listed, or approved for listing, on the applicable qualified products list for the engine-airframe combination specified in Table I; and which is in compliance with this specification.

3.2 Performance reliability.—Spark plug shall be capable of operating satisfactorily without fouling or other evidence of electrical or mechanical failure when subjected to the flight tests specified in 4.3.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling.—

4.1.1 Acceptance Inspections shall be in accordance with MIL-S-7886. The Sampling Plan for Group B inspection shall be in accordance with MIL-STD-414, Method B, C, or D; AQL-1.0% for the Lower Limit and AQL-2.5% for the Upper Limit, Level I. Lot size not to exceed one continuous production run of a specific design for a specific contract. Contractor shall be permitted to re-establish qualification limits at any time that repairs or modifications to test engines bias rating data.

4.2 Test conditions.—

4.2.1 Engine.—Only an engine in proper mechanical and electrical condition, having sufficient time before the next overhaul for the flight test involved, shall be used for the testing of the spark plugs. Bushing threads in the cylinder shall be cleaned of carbon and other foreign matter. The maintenance inspection of the engine shall also include a check on (1) the compression and (2) the ignition components including coils, breaker points, condenser, ignition harness, distributor, and magneto timing.

4.2.2 Identification of plugs.—Prior to installation of each complete set of test plugs, of the same part number, each plug shall be permanently marked by etching to indicate cylinder number and position.

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4.2.3 Gasket.--Whenever a plug is installed, a new gasket conforming to MS17840 shall be used.

4.2.4 Torque.--Plugs shall be tightened to the torque specified for the test condition, preferably to the high limits, as specified in MIL-S-7886.

4.2.5 Lubrication.--Whenever a plug is removed for inspection, the threads shall be lubricated with engine oil conforming to grade 1010 of MIL-L-6081, prior to its reinstallation.

4.3 Tests.--Spark plugs shall be tested under controlled, and under other than controlled flight conditions, as specified below. Engine-airframe combinations, fuel, electrode gap, flight hours, and allowable number of spark plug failures for each engine, shall be as specified in Table I.

TABLE I. — Engine-Airframe Combinations

Engine	Aircraft	Fuel	Electrode gap (inch)	Flight hours	Plug failures per engin
R-1340-59, -61	U-1A	80/87	.015/.018	400	4
O-470-11, -15	O-1A, -1D, -1E, -1G	80/87	.015/.018	400	3
R-985-AN-39A	U-6A	80/87	.015/.018	400	4
O-480-1A, -1B -3, and -3A	U-8D, U-8F, U-8G	115/145	.015/.018	400	3
O-435-23C	OH-13H	80/87	.015/.018	400	3

4.3.1 Controlled flight.--At the beginning of the controlled flight tests, and at approximately every 100 hours of flight time thereafter, the following three tests, for both rotary wing and fixed wing, shall be successfully accomplished three times:

4.3.1.1 Rotary wing.--

4.3.1.1.1 Lifting and hovering capability.--Engine shall be allowed to idle in auto rich for 15 minutes, using normal defouling procedures. Engine shall then be capable of lifting and hovering the craft with its maximum military load until highest permissible engine cylinder head temperature is reached.

4.3.1.1.2 Maximum rate of climb and power.--After engine cools to normal operating temperature, it shall be capable of taking off and maintaining maximum rate of climb to the aircraft maximum operating

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altitude, followed by a 15-minute cruise at low power to cool and stabilize engine temperatures, then a level course maintained at maximum except takeoff (METO) for 5 minutes, followed by maximum endurance power for 30 minutes.

4.3.1.1.3 Cruise power and autorotation.--Engine shall be cruised at low power for 15 minutes, to cool the engine, and autorotated to a safe recovery altitude, followed by level flight at cruise power for 30 minutes.

4.3.1.2 Fixed wing.--

4.3.1.2.1 Maximum rate of climb with maximum military load.--Engine shall be allowed to idle in auto rich for 15 minutes, using normal defouling procedures. Engine shall then be capable of maintaining maximum rate of climb with maximum military load from point of takeoff to the aircraft maximum operating altitude.

4.3.1.2.2 Cruise and maximum endurance power.--The engine shall then be operated for 15 minutes at approximately 65-percent power in cruise to cool and stabilize engine temperatures, followed by maximum except takeoff (METO) power for 5 minutes, and then at maximum endurance power for 30 minutes.

4.3.1.2.3 Cruise, glide, and level flight.--Then cruise shall be at approximately 65-percent power for another 15 minutes to cool the engine, followed by a 2-minute glide at 20-percent power, and again at approximately 65-percent power in level flight for 15 minutes.

4.3.2 Other than controlled flight.--Flight time other than controlled shall be under normal service conditions and in accordance with applicable technical data relating to performance of typical maneuvers normally associated with observation, resupply, communication relay, and similar type missions.

4.4 Inspections.--

4.4.1 Intervals.--Plugs shall be tested without their removal by using an ignition analyzer at 50, 100, 150, 200, 250, 300, 350, and 400 hours, to determine whether the plugs in each cylinder function properly.

4.4.1.1 Malfunctioning plugs.--Any plug showing indication of malfunctioning shall be removed from the engine and visually inspected. If oil-fouled, the plugs shall be cleaned by soaking the firing end cavity for 30 minutes in solvent conforming to P-D-680, then draining and

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thoroughly drying them with a filtered airblast. A light abrasive blast cleaning is permissible if necessary to remove hardened oil deposits. In addition, the ceramic shielding barrel shall be thoroughly cleaned. The electrode gaps shall be measured with a round wire gage. (Plugs shall not be regapped.) Plug shall be placed in a spark plug tester such as the AV 18-1 manufactured by AC Spark Plug Div. of General Motors Corp., or equivalent. Tester shall be connected to a single-phase 110-120 volt, 60-cycle power supply outlet. Nitrogen gas shall be used in the pressure chamber: for spark plug electrode gaps under .020-inch, 200 psi; for gaps over .020-inch, 175 psi. If plug fires under these conditions, it shall be reinstalled in same position in the engine, using a new gasket conforming to MS17840.

4.4.1.2 Plug failures.--Plugs other than oil-fouled, or showing obvious mechanical failure, shall be tested in a spark plug tester (see 4.4.1.1). Any plug failing to fire shall be considered a failed plug. Plug failures due to other than normal flight test conditions are not to be considered plug failures and shall be replaced with like new items. Cracked ceramic insulators, electrode failures, chronic fuel and lead fouling, or other bonafide failures are considered plug failures.

4.4.2 Final examination.--At the completion of 400 flight hours, plugs shall be removed from the engine and given final inspection. This shall include visual inspection, final gap measurement, and testing in spark plug tester as specified in 4.4.1.1. Results of these inspections shall be shown on test report indicating the performance of each set of spark plugs under tests and examinations specified herein.

4.5 Test stoppage.--If more than the allowable number of plug failures occurs before completion of flight hours scheduled (see Table I), test shall be discontinued.

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery.--Preparation for delivery shall be in accordance with Section 5 of MIL-S-7886.

6. NOTES

6.1 Intended use.--Type 4 spark plug is intended for use with the engine-airframe combinations specified in 4.3.

6.2 Ordering data.--Ordering data shall be in accordance with MIL-S-7886.

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6.3 Qualification.--With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Headquarters, U. S. Army Aviation Systems Command, ATTN: AMSAV-EK, PO Box 209, Main Office, St. Louis, Missouri 63166, and information pertaining to qualification of products may be obtained from that activity.

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